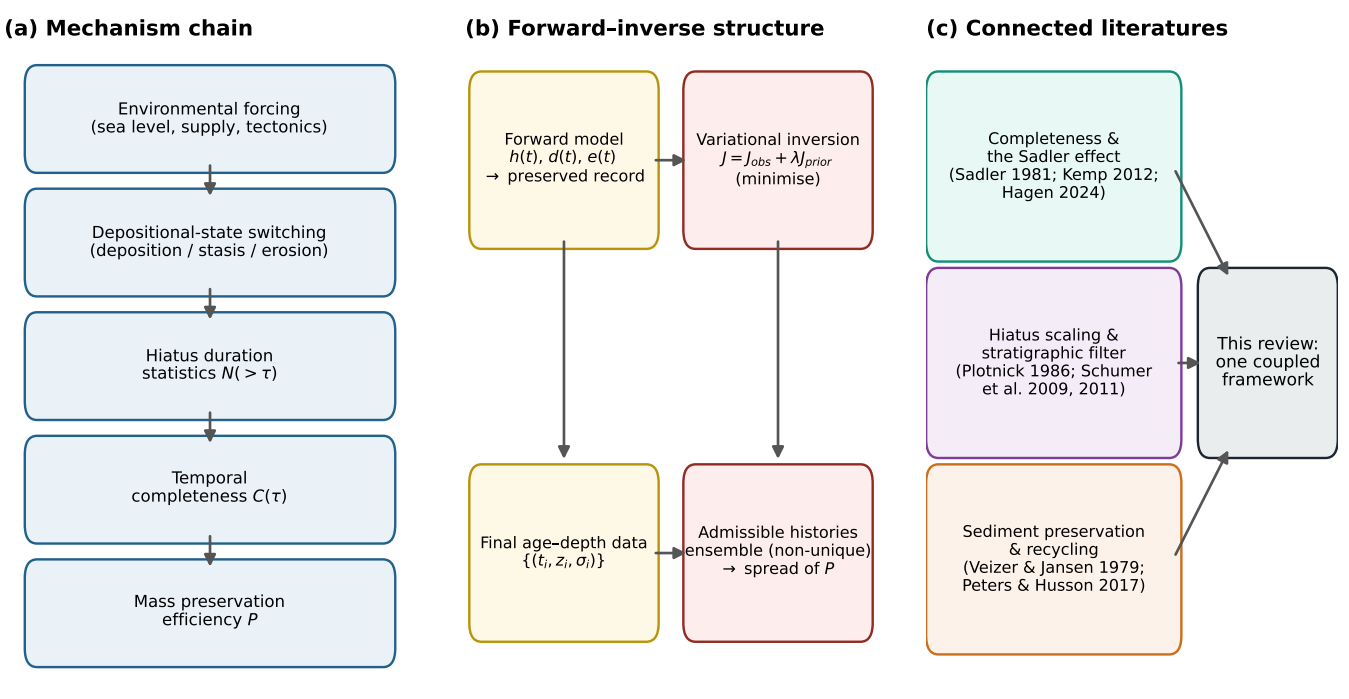
(a) Mechanism chain
Environmental forcing
(sea level, supply, tectonics)
Depositional-state switching
(deposition / stasis / erosion)
Hiatus duration
statistics $N(>\tau)$
Temporal
completeness $C(\tau)$
Mass preservation
efficiency P
(b) Forward–inverse structure
Forward model
$h(t)$, $d(t)$, $e(t)$
→ preserved record
Variational inversion
$J = J_{obs} + \lambda J_{prior}$
(minimise)
Final age–depth data
$\{(t_i, z_i, \sigma_i)\}$
Admissible histories
ensemble (non-unique)
→ spread of P
(c) Connected literatures
Completeness &
the Sadler effect
(Sadler 1981; Kemp 2012;
Hagen 2024)
Hiatus scaling &
stratigraphic filter
(Plotnick 1986; Schumer
et al. 2009, 2011)
Sediment preservation
& recycling
(Veizer & Jansen 1979;
Peters & Husson 2017)
This review:
one coupled
framework